



SLEEP-Wake Disorders

Introduction

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- ❑ Normal humans sleep up to **one-third** of their lives and spend more time sleeping compared with any other single activity
- ❑ Optimal sleep duration is 7–9 hours/24-hour period.
- ❑ Sleep is a reversible state of **unresponsiveness to stimuli** of the outside world
 - ❑ Sleep is necessary to maintain **wakefulness, health, and welfare.**

Introduction

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- Unfortunately, sleep disorders are prevalent & represent a major cause of *societal morbidity, lost productivity, and reduced QOL*.
 - ▣ **Sleep disturbances** may contribute to the **development** and progression of comorbid medical conditions
- Approximately 50% of adults will report a sleep complaint over the course of their lives
- In Ethiopia 53% of the population had poor sleep quality, ranging from 26 to 66.2%.

Physiology Recap

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- **Sleep** is governed and paced by the **suprachiasmatic nucleus** in the brain that regulates **circadian rhythm**
- Sleep is divided into two phases
 - ▣ Non-rapid eye movement (NREM)
 - ▣ Rapid eye movement (REM)
- There are four stages of NREM
 - ▣ Healthy sleep will typically progress through the four stages of NREM sleep prior to the REM period

NREM

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□ Stage 1

- It is the first stage of sleep (5% of sleep)
- A brief period of transition between wakefulness and sleep
- Breathing is irregular
- Characterized by theta wave in EEG
- individuals describe this stage as being awake, being drowsy, or as being asleep

Relaxed wakefulness

Alpha waves



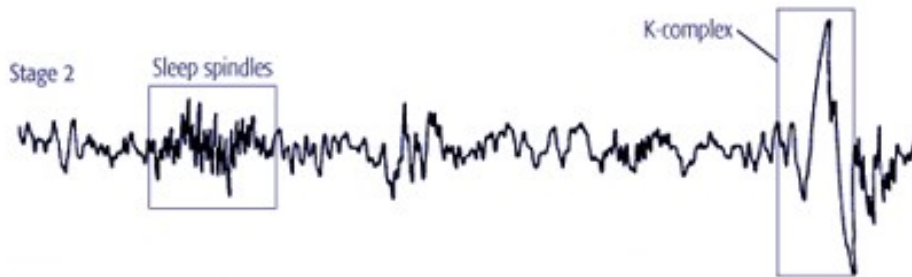
Stage 1

Theta waves



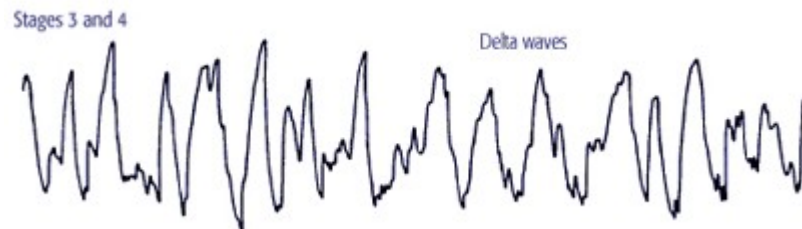
□ Stage 2

- First true Sleep State
- Accounts for 50% of total sleep time
- Breathing and heart rate become regular and body temperature starts to go down
- EEG: sleep spindles, K complexes

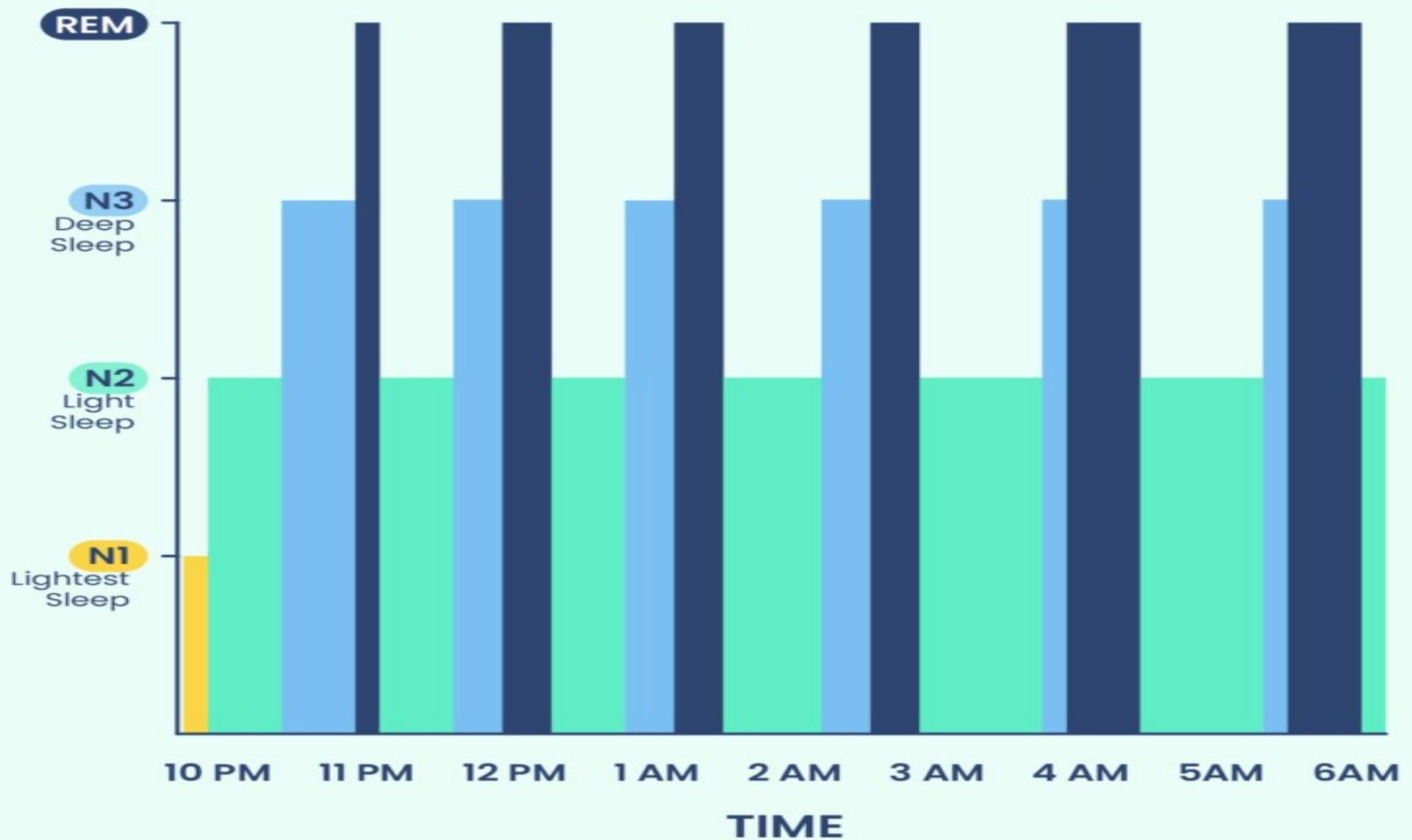


□ Stage 3 and 4

- Slow metabolic activity and brain waves
- Blood supply to the muscles is increased, energy is restored, tissue growth and repair occur.
 - Restorative sleep
- Breathing is slow and regular
- Blood pressure and body temp drop
- Increasing proportion of Delta (slow) wave on the EEG



Sleep Cycles Through the Night



Features of NREM sleep

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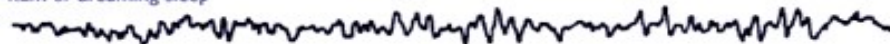
- ❑ Metabolic activity and brain wave slow
- ❑ Reduced recall of dreaming if woken
- ❑ Reduced complexity of dreams
- ❑ Increased parasympathetic activity
- ❑ Few or no eye movements
- ❑ Abolition of tendon reflexes
- ❑ Decreased heart rate
- ❑ Decreased systolic blood pressure
- ❑ Decreased respiratory rate
- ❑ Decreased cerebral blood flow
- ❑ Penis not usually erect

REM sleep

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- Dramatic physiological change
- The brain becomes electrically and metabolically activated.
- It is accompanied by a 62% to 173% increase in cerebral blood flow
- Generalized muscle atonia
- Bilateral rapid eye movements
- Dreaming
- Fluctuations in respiratory and cardiac rate
- EEG resembles wakefulness (fast)

REM or dreaming sleep

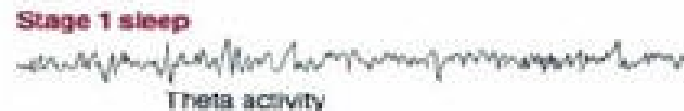


Awake



desynchronized

Stage 1



Stage 2



Stage 3

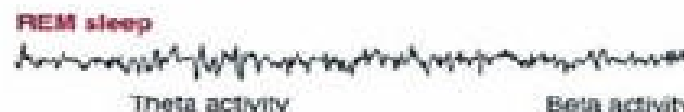


Stage 4



Slow-wave sleep,
synchronized
(delta waves).

REM



desynchronized

Features of REM sleep

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- Increased recall of dreams if woken
- Increased complexity of dreams
- Increased sympathetic activity
- Transient runs of conjugate eye movements
- Maximal loss of muscle tone
- Increased heart rate
- Increased systolic blood pressure
- Increased respiratory rate
- Increased cerebral blood flow
- Occasional myoclonic jerks
- Penile erection or increased vaginal blood flow

Primary sleep disorders***Dyssomnias—abnormality in the amount, quality, or timing of sleep***

Primary insomnia

Primary hypersomnia

Narcolepsy

Breathing-related sleep disorder

Circadian rhythm sleep disorder

 Delayed sleep phase type

 Jet lag type

 Shift work type

 Unspecified type

Dyssomnia not otherwise specified

Parasomnias—abnormal behavioral or physiologic events associated with sleep

Nightmare disorder

Sleep terror disorder

Sleepwalking disorder

Parasomnia not otherwise specified

Sleep disorders related to another mental disorder

Insomnia or hypersomnia related to another mental disorder

Dyssomnias: are primary disorders of initiating or maintaining sleep or of excessive sleepiness

are characterized by a disturbance in the **amount, quality, or timing of sleep**

- **Insomnia** is most frequently a symptom or manifestation of an underlying disorder (**2^o insomnia**) but may occur in the absence of contributing factors (**1^o insomnia**).
- It is difficulty of falling asleep (**>30 mins**), maintaining sleep, **or not feeling rested** in spite of a sufficient opportunity to sleep.

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- *Resulting daytime sleepiness & fatigue are troubling after-effects.*
 - *Early treatment of insomnia may prevent the development of psychopathologic complications.*
 - **Parasomnias** are disruptive sleep disorders that can occur during arousals from sleep or arousals from non-rapid eye movement (NREM) sleep.
 - They can result in undesirable physical or verbal behaviors, like **walking or talking during sleep.**



Insomnia

Cont.....

- Insomnia is the most common complaint in general medical practice, occurring in **30 – 45%** of adults.
- Patients with insomnia complain of difficulty falling asleep, maintaining sleep, or experiencing **nonrestorative** sleep
- It causes distress, due to fear of unable to fall asleep at bedtime and
- can impair work-related productivity because of daytime fatigue or drowsiness

Cont.....

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- Persons with insomnia are **10 times** more likely to suffer from **depression** & **17 times** more likely to have **anxiety** compared with people without insomnia.
- **Chronic insomnia** is 2-3 times greater in individuals with **high BP, breathing difficulty, GI disorders, cancer & chronic pain**, among other conditions.

Clinical Presentation

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- **Transient** (2 to 3 nights) and **short-term** (<4 wks) insomnia is common & is usually related to a **precipitating factor**.
- **Chronic insomnia** (>1 mth) may be related to medical or psychiatric disorders or medication, or it may be psychophysiologic.

Common Etiologies of Insomnia

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Situational

Work or financial stress, major life events, interpersonal conflicts
Jet lag or shift work

Medical

Cardiovascular (angina, arrhythmias, heart failure)
Respiratory (asthma, sleep apnea)
Chronic pain
Endocrine disorders (diabetes, hyperthyroidism)
GI (gastroesophageal reflux disease, ulcers)
Neurologic (delirium, epilepsy, Parkinson's disease)
Pregnancy

Psychiatric

Mood disorders (depression, mania)
Anxiety disorders (e.g., generalized anxiety disorder, obsessive-compulsive disorder)
Substance abuse (alcohol or sedative-hypnotic withdrawal)

Pharmacologically induced

Anticonvulsants
Central adrenergic blockers
Diuretics
Selective serotonin reuptake inhibitors
Steroids
Stimulants

Diagnosis

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- In pts with chronic disturbances, a diagnostic evaluation includes:
 - Physical & mental status examinations,
 - Routine laboratory tests, and
 - Medication & substance abuse histories.
- **One month** of symptoms is required before a sleep disorder is diagnosed.

Treatment

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- General Management includes:
 - **Identifying the cause** of insomnia,
 - Education on sleep hygiene,
 - Stress management,
 - Monitoring for mood symptoms
- Transient & short-term insomnia should be treated with **good sleep hygiene** & careful use of sedative -hypnotics if necessary.

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- **Chronic insomnia** calls for
 - ☑ Careful assessment for a medical cause
 - ☑ Non-pharmacologic treatment and
 - ☑ Careful use of sedative-hypnotics (intermittently to prevent tolerance and dependence).

Non- Pharmacologic Therapy

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- Behavioral and educational interventions that may help include
 - Short-term cognitive behavioral therapy (CBT),
 - Stimulus control therapy
 - Sleep restriction
 - Sleep hygiene education

Recommendations for Insomnia

Stimulus control procedures

1. Establish regular times to wake up and to go to sleep (including weekends).
2. Sleep only as much as necessary to feel rested.
3. Go to bed only when sleepy. Avoid long periods of wakefulness in bed. Use the bed only for sleep or intimacy; do not read or watch television in bed.
4. Avoid trying to force sleep; if you do not fall asleep within 20–30 minutes, leave the bed and perform a relaxing activity (e.g., read, listen to music, or watch television) until drowsy. Repeat this as often as necessary.
5. Avoid daytime naps.
6. Schedule worry time during the day. Do not take your troubles to bed.

Sleep hygiene recommendations

1. Exercise routinely (three to four times weekly) but not close to bedtime because this can increase wakefulness.
2. Create a comfortable sleep environment by avoiding temperature extremes, loud noises, and illuminated clocks in the bedroom.
3. Discontinue or reduce the use of alcohol, caffeine, and nicotine.
4. Avoid drinking large quantities of liquids in the evening to prevent nighttime trips to the restroom.
5. Do something relaxing and enjoyable before bedtime.

Non-benzodiazepine Hypnotics

- ✓ **First generation antihistamines**
(diphenhydramine ,doxylamine, and pyrilamine) are less effective than BZs, but side effects are usually minimal.
- ✓ Their anticholinergic side effects may be problematic, especially in the elderly.

- **Antidepressants** are good alternatives for pts who should not receive BZs, especially those with depression or a history of substance abuse.
 - **Amitriptyline 25 mg at night, doxepin, and nortriptyline** are effective, but side effects include anticholinergic effects, adrenergic blockade, and cardiac conduction prolongation.

Cont.....

- **Trazodone**, 25 to 100 mg, is often used for insomnia induced by SSRIs or bupropion.
- At these low doses, it induces and maintains sleep without causing daytime drowsiness due to its short half-life (3–6 hours)
- But for Antidepressant effects simultaneous blocking of 5-HT_{2A} is required, which occurs at higher doses (150–600mg)

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- **Ramelteon** 8 mg at bedtime is a **melatonin receptor** agonist selective for the MT1 & MT2 receptors.
- Melatonin is **not a sedative**, but a sign of darkness or night-time, and is a cue for innate night-time behaviour.
- In the morning when plasma levels of melatonin decrease, sleep ends and wakefulness begins
- $T_{1/2}$ of 1.5 - 5 hr. Dose of up to 160mg (unscheduled dosing)
- It is effective in pts who have **sleep onset insomnia**.
- Common side effects: headache, somnolence & sore throat.

Cont.....

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- **Zolpidem** acts selectively on GABA_A and has minimal anxiolytic and no muscle relaxant or anticonvulsant effects ($t_{1/2} = 1.5-2.4\text{hr}$)
- It is indicated for the short-term treatment of difficulty **sleep initiation insomnia**
- most common side effects are headache, dizziness, and somnolence.
- Zolpidem is not approved for long-term use

Cont.....

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- **Zaleplon** also binds to the GABA_A receptor.
- It is very rapid acting ($t_{1/2}$,1hr).
- Used for pts who have **difficulty of falling sleep**.
- persistent sleepiness and rebound insomnia **is not** associated with zaleplon
 - Occasional side effects: headache, dizziness, nausea, abdominal pain and somnolence
 - Is not indicated for long term use

Benzodiazepines

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- **Triazolam** is distributed quickly and has short duration of effect.
 - Erythromycin, nefazodone, fluvoxamine, and ketoconazole **reduce the clearance** of triazolam and increase plasma concentrations.
- **Estazolam** and **temazepam** are intermediate in their duration of action.
- The effects of **flurazepam** and **quazepam** are long because of **active metabolites**.

PKs of hypnotic

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Generic Name (Brand Name)	t_{max} (hours) ^a	Half-Life ^b (hours)	Daily Dose Range (mg)	Metabolic Pathway	Clinically Significant Metabolites
Estazolam (ProSom)	2	12–15	1–2	Oxidation	—
Eszopiclone (Lunesta)	1–1.5	6	2–3	Oxidation Demethylation	—
Flurazepam (Dalmane)	1	8	15–30	Oxidation	Hydroxyethylflurazepam, flurazepam aldehyde
Quazepam (Doral)	2	39	7.5–15	<i>N</i> -dealkylation Oxidation, <i>N</i> - dealkylation	<i>N</i> -desalkylflurazepam ^c 2-Oxo-quazepam, <i>N</i> - desalkylflurazepam ^c
Temazepam (Restoril)	1.5	10–15	15–30	Conjugation	—
Triazolam (Halcion)	1	2	0.125–0.25	Oxidation	—
Zaleplon (Sonata)	1	1	5–10	Oxidation	—
Zolpidem (Ambien)	1.6	2–2.6	5–10	Oxidation	—

- All BZs, except **temazepam** are metabolized by microsomal oxidation followed by glucuronide conjugation to be eliminated.
- **Side effect of BZs:**
 - Drowsiness, psychomotor incoordination, decreased concentration, and cognitive deficits.
 - **Tolerance to the daytime CNS effects** may develop in some individuals.
 - **Tolerance to hypnotic effects** develops after 2 wks

Side effects...

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- Residual daytime sedation
- Rebound insomnia
- Anterograde amnesia
- Withdrawal symptoms

Case – insomnia

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1. A.M, a 31-year-old woman, is requesting a medication for treatment of her insomnia. She just returned to Addis from US 2 days ago and she is now having difficulty getting to sleep. When she arrived Addis, she went to sleep immediately at 4 PM and awoke at 3 AM. it now takes her 2 to 3 hours to fall asleep. She has difficulty awakening in the morning and, as a result, sleeps past noon. She needs to be alert during the day to fulfill her obligations.

- What information provided by A.M. is important in the assessment of her insomnia?
- what counseling is needed for her?
- What appropriate therapy do you recommend for her?

2. A 38-year-old kindergarten teacher presents to the clinic today with noticeable dark circles under the eyes. She has difficulty with sleep, mainly with staying asleep. It takes about 20 minutes to fall asleep, but after about 5 hours, she wakes up and cannot fall asleep again for several hours. This pattern has negatively affected her job performance, and she feels tired all the time. She once took diphenhydramine for sleep, but had to miss work because of extreme drowsiness in the morning. They wonder whether they could take any other medications. Other medical problems include hypothyroidism (levothyroxine 125 mcg at bedtime), hypertension (hydrochlorothiazide 25 mg in the morning), chronic back pain (ibuprofen 800 mg three times daily), and MDD (citalopram 20 mg in the morning).

2.1. Which agent is most likely contributing to insomnia?

- A. Citalopram.
- B. Hydrochlorothiazide.
- C. Ibuprofen.
- D. Levothyroxine.

2.2. Which medication used for insomnia is most appropriate to recommend for this patient after adjusting the medication in question?

- A. Trazodone.
- B. Temazepam.
- C. Zaleplon.
- D. Zolpidem

The End!